GMC Radiation Monitor

Local radiation monitoring for compatible GQ GMC Geiger counters

User manual
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Important safety notice

The app is a monitoring, analysis and home-automation tool. It does not replace a calibrated radiation-protection instrument, an official measurement or a professional risk assessment. Unusual or persistently high readings require the measurement setup, device and environment to be checked and, where appropriate, competent authorities or specialists to be consulted.

Measurement	CPM; derived µSv/h only with an appropriate factor
Storage	Local SQLite database
Long-term windows	24 h, 7 d, 30 d, 90 d, 365 d
Languages	German, English, Spanish, French, Croatian, Italian, Dutch, Polish

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1. Purpose and system overview - Overview

- GMC Radiation Monitor reads compatible GQ GMC Geiger counters locally over USB serial, stores confirmed measurements in SQLite and publishes Home Assistant entities through MQTT Discovery.
- The interface separates live status, analysis and expert information. The new long-term analysis complements the existing real-time alerts, adaptive background and multi-device analysis without duplicating them.

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1. Purpose and system overview - Practice and verification points

- Measurements, reports and backups remain local. External transfer occurs only through explicitly enabled functions such as GMCMap.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

2. Installation and first start - Overview

- Install the app from the Home Assistant app repository, verify that MQTT is available and start the app after saving the device configuration.
- Give every counter a unique name and, where possible, bind it to its serial number. Select the correct detector profile; custom conversion factors should have a documented source.

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2. Installation and first start - Practice and verification points

- Open the web interface and check device status, data age, CPM, dose rate and stored history. A newly connected device may need a short synchronisation period.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

3. Navigation and information levels - Overview

- Summary shows the essential live values, connection states, warnings and plain-language conclusions.
- Analysis shows background comparison, trends, long-term windows, events, environmental context and device comparisons.

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3. Navigation and information levels - Practice and verification points

- Expert view shows raw-data quality, detector factors, dead-time correction, statistical diagnostics and technical protocol information.
- Collapsible groups prevent overload. On small screens, wide tables scroll inside their card while the page itself remains free of horizontal overflow.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

4. Live measurements, alerts and data quality - Overview

- CPM is the primary count rate. A µSv/h value is a device- and detector-dependent derivation and is only as reliable as the conversion factor and calibration.
- Absolute alarm thresholds and smart notifications remain separate from long-term statistics. Statistical signals never alter radiation-safety thresholds.

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4. Live measurements, alerts and data quality - Practice and verification points

- Unconfirmed isolated spikes are not stored as confirmed history. Quality flags, gaps, timestamps and device availability remain traceable.
- With multiple devices, each counter is monitored independently; one offline device does not set the others offline.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

5. Long-term analysis: windows and background - Overview

- The app calculates real 24-hour, 7-day, 30-day, 90-day and 365-day windows rather than merely taking the last N records.
- A number is shown only when duration and minimum coverage are sufficient. Otherwise the card states “Not yet meaningful” and explains why.

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5. Long-term analysis: windows and background - Practice and verification points

- Missing values are not interpolated. Mean, median, quantiles and coverage use only existing accepted measurements.
- A robust local background is calculated per device and context. Connected relative elevations are grouped into episodes with start, duration, mean, maximum and excess area.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

6. Cumulative dose and development over time - Overview

- Cumulative dose is integrated from stored dose-rate values. It is a derived estimate, not personal dosimetry.
- An annual projection is shown only from a sufficiently long and complete basis and is clearly labelled as a projection.

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6. Cumulative dose and development over time - Practice and verification points

- Daily and monthly values, calendar heat map, rolling seven-day median and month-of-year profile reveal long-term development. The seasonal profile requires at least six represented calendar months.
- Changes in location, geometry, detector, firmware or calibration can influence long-term patterns and must be considered.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

7. Advanced statistics - Overview

- Effective sample size accounts for temporal dependence between consecutive measurements.
- Moving-block bootstrap provides robust confidence intervals for time-dependent data. Mann-Kendall and Sen slope describe monotonic trends without assuming normality.

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7. Advanced statistics - Practice and verification points

- EWMA and CUSUM can reveal small sustained level shifts. Overdispersion describes variability beyond a simple count model.
- These methods provide context only. They do not identify a radiation source or prove a cause.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

8. Environmental, event and intervention analysis - Overview

- Optional Home Assistant temperature and pressure sensors can be linked to the radiation series.
- Lagged Spearman associations are explored at 0, 3, 6, 12 and 24 hours. Correlation does not imply causation.

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8. Environmental, event and intervention analysis - Practice and verification points

- Event annotations document moves, weather events, building changes or checks. They appear in reports and support before/after review.
- Comparisons should use equivalent device, location, detector and timing conditions wherever possible.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

9. Multi-device comparison and agreement - Overview

- The existing fleet view compares time-paired values and shows shared rises, relative calibration, drift and position changes.
- Bland-Altman bias and 95% limits of agreement complement correlation. High correlation alone does not demonstrate agreement.

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9. Multi-device comparison and agreement - Practice and verification points

- Comparisons between different tubes or conversion factors must remain device-specific. CPM and µSv/h are not interchangeable without evidence.
- Unusual differences should be checked with parallel measurements, setup documentation and calibration records.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

10. Reports, exports and GMCMap - Overview

- Reports can be generated by device, period and display mode. CSV and JSON support further analysis; PDF reports document results and limitations.
- Long series are peak-preservingly reduced for charts while statistics and data exports continue to use the relevant full dataset.

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10. Reports, exports and GMCMap - Practice and verification points

- GMCMap is optional. Enable upload only after privacy confirmation and enter the account and device-specific counter ID correctly.
- A GMCMap upload is not a replacement for local storage or a scientific archive.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

11. Backup, restore and deletion - Overview

- Managed SQLite backups and full exports protect the history. Test backups and retain important copies outside the Home Assistant system.
- Restore and complete deletion are disabled by default. Enable history_management_enabled only for planned maintenance and disable it afterwards.

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11. Backup, restore and deletion - Practice and verification points

- Restores are checked for schema and SQLite integrity before data are merged. Deletion requires explicit text confirmation and is protected server-side.
- Home Assistant Recorder history and the local app database are separate storage areas.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

12. Configuration and maintenance - Overview

- Important groups are devices, GMCMap, history, analysis, safety, interface and system. Change only parameters whose effect is understood.
- For annual windows, retention must be at least 365 days; new installations default to 730 days.

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12. Configuration and maintenance - Practice and verification points

- After updates, verify the displayed version, device assignments, conversion factors, MQTT entities and report language.
- Review diagnostics and logs for location, serial-number and account information before sharing them.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

13. Troubleshooting and scientific limits - Overview

- No connection: check USB assignment, cable, model, startup delay and app log. Use diagnostics for fragmented serial replies instead of arbitrarily shortening timing values.
- No long-term values: check duration, coverage, retention and device filters.

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13. Troubleshooting and scientific limits - Practice and verification points

- Implausible dose rate: check detector profile, CPM-per-µSv/h factor, dead-time model and calibration reference.
- Statistical sophistication cannot replace calibration, suitable placement or knowledge of energy response. Radiation-protection decisions require competent expertise.

Practice and verification points	For relevant changes, record the device, detector profile, location, time and reason.

Appendix A - Quick reference

Measurement	CPM; derived µSv/h only with an appropriate factor
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Overview

Long-term values appear only with sufficient duration and coverage. Missing measurements are not interpolated.

Appendix B - Terms and methods

CPM	Counts per minute; the primary device count rate.
µSv/h	Derived dose rate; depends on detector, energy response, factor and calibration.
Data coverage	Share of a time window represented by accepted measurements.
Effective sample size	Estimates the number of statistically independent observations under autocorrelation.
Block bootstrap	Resamples connected time blocks to produce robust confidence intervals.
Sen slope	Robust estimate of a monotonic change per day.
EWMA / CUSUM	Contextual methods for detecting small sustained level changes.
Bland-Altman	Assesses bias and limits of agreement between two time-paired devices.

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Simplified long-term presentation

Long-term analysis now starts with a plain-language assessment. Method details remain available in collapsible sections. The overview focuses on trend, background context, data basis, real time windows and integrated derived dose.

- Advanced statistics are collapsed by default.
- Consistent type sizes and responsive cards.
- Unavailable results are explained instead of shown as an unexplained dash.

GMC Radiation Monitor is a monitoring and analysis tool. Statistical results do not replace calibrated radiation-protection measurements or professional assessment.

Evidence levels and minimum requirements

Statistical results are labelled not evaluable, exploratory, preliminary, supported or well supported. The level uses duration, coverage and effective sample size.

- 24 hours: at least 90% coverage.
- Long-term trend: at least 30 sufficiently complete days.
- Environmental associations: at least 100 hourly pairs per lag.
- Annual projection: only after a complete 90-day window.

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Practical effect size and dose wording

A p-value alone does not show practical importance. The app therefore adds estimated monthly change and a practical magnitude. Dose terms are explicitly separated.

- Integrated derived dose: accumulated over the measured period.
- Annual projection: a calculation assuming an unchanged level.
- Without a documented factor, presentation remains in CPM.

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Environmental associations and interpretation

Lagged Spearman associations with temperature and pressure are exploratory. Benjamini-Hochberg correction reduces false discoveries across tested lags. Correlation does not establish causation.

- Pair count, lag and FDR-adjusted p-value are shown.
- EWMA, CUSUM and structural signals do not identify a source.
- Statistical models never suppress configured alarms.

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Field-test protocol and long-term diagnostics

The expert view contains operational counters for practical endurance testing. A JSON protocol can be exported and archived with measurement reports.

- Runtime and reconnects.
- Serial errors, database write errors and longest gap.
- GMCMap successes and failures.
- Diagnostics do not certify calibration or measurement accuracy.

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